

Lesson 4 - Cleaning the Data

READ ME FIRST (theory). Then open the practice notebook.

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Lesson 4 - Cleaning the Data

Real data is dirty. The model needs clean data. The theory first, then `practice.ipynb`.

Key words

Word	Simple meaning
missing value (NaN)	an empty cell
dtype	the type of a column (number or text)
median	the middle value of a column

Word	Simple meaning
mode	the most common value
outlier	a strange value, very far from the rest

1. Why clean the data?

“Bad data in, bad model out.” A model cannot learn from text where it expects numbers, or from empty cells. We fix these first.

2. A column that looks numeric but is text

Sometimes a column like `TotalCharges` looks like numbers but pandas reads it as **text** (object). Reason: a few cells are blank spaces, not numbers. One bad cell makes the whole column text.

Fix it:

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"], errors="coerce")
```

- `to_numeric` turns text into numbers.
- `errors="coerce"` = if a cell is not a number, make it **NaN** (instead of crashing).

Common mistake: doing math on the column before fixing the type -> error.

3. Find missing values

```
df.isna().sum().sort_values(ascending=False)
```

This shows how many empty cells are in each column.

4. Fill missing numbers - median

Fill empty number cells with the **median** (the middle value):

```
m = df["TotalCharges"].median()
df["TotalCharges"] = df["TotalCharges"].fillna(m)
```

Why median, not mean? A few very big values pull the mean up. The median is not moved by a few outliers, so it is safer.

5. Fill missing text - mode

For a text column, fill with the **mode** (the most common value):

```
top = df["Contract"].mode()[0]
df["Contract"] = df["Contract"].fillna(top)
```

6. Duplicates

Two identical rows can confuse the model.

```
df.duplicated().sum()      # how many duplicate rows
df = df.drop_duplicates()  # remove them
```

7. Unrealistic values (outliers)

Check that numbers make sense. A tenure of 999 months or a negative charge is wrong. Use `.describe()` to see the min and max, then decide to remove or cap them.

Why: crazy values pull the model in the wrong direction.

Quick summary

- Fix wrong **types** first (`to_numeric(..., errors="coerce")`).
- Find missing with `df.isna().sum()`.
- Fill numbers with **median**, text with **mode**.
- Remove **duplicates**. Check for **unrealistic** values with `.describe()`.

Now do this

Open `practice.ipynb`, clean the real Telco data, and do the assignment.